Since **Unit III** covers **Data Preprocessing, Association Rule Mining (Apriori, FP-Growth), KDD, and Data Visualization**, and **Unit IV** focuses on **Classification and Prediction**, the mini-projects should enable students to perform the complete data mining pipeline: **data collection → preprocessing → visualization → model building → evaluation** using Python and Scikit-learn.

JURY Statements (Units III & IV)

1. Customer Purchase Pattern Analysis and Product Recommendation

Problem Statement:
Develop a data mining application to analyze retail transaction data using Apriori and FP-Growth algorithms. Identify frequent itemsets and association rules to recommend products frequently purchased together and visualize customer purchasing behavior.

Concepts Covered

- KDD Process
- Data Preprocessing
- Association Rule Mining
- Apriori
- FP-Growth
- Data Visualization

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2. Student Academic Performance Prediction

Problem Statement:
Develop a predictive analytics model to forecast students' academic performance based on attendance, internal assessments, assignment submissions, and participation. Compare multiple classification algorithms to identify the best-performing model.

Concepts Covered

- Data Cleaning
- Feature Engineering
- Decision Tree
- Naïve Bayes
- KNN

- Random Forest
- Model Evaluation

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3. Credit Card Fraud Detection using Classification Techniques

Problem Statement:
Design a fraud detection system that analyzes credit card transaction data to classify transactions as legitimate or fraudulent. Perform data preprocessing, handle class imbalance, compare classification algorithms, and evaluate model performance using suitable metrics.

Concepts Covered

- Data Preprocessing
- Classification
- SVM
- Random Forest
- Gradient Boosting
- Precision, Recall, F1-score

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4. Customer Churn Prediction for Subscription Services

Problem Statement:
Build a predictive model to identify customers likely to discontinue a subscription service. Apply preprocessing techniques, compare multiple classification algorithms, and determine the most influential factors affecting customer retention.

Concepts Covered

- KDD Process
- Data Cleaning
- Decision Tree
- Naïve Bayes
- Random Forest
- Feature Importance

- Model Evaluation

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5. Disease Risk Prediction using Healthcare Data

Problem Statement:
Develop a classification model to predict the risk of diseases such as diabetes or heart disease using patient demographic and clinical data. Compare machine learning algorithms and analyze their prediction accuracy.

Concepts Covered

- Data Preprocessing
- Data Visualization
- Decision Tree
- SVM
- KNN
- Random Forest
- Performance Evaluation

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6. Email Spam Detection System

Problem Statement:
Develop an intelligent email classification system that categorizes emails as spam or legitimate using text-based features. Perform preprocessing, transform textual data into machine-readable format, and evaluate multiple classification algorithms.

Concepts Covered

- Data Cleaning
- Feature Extraction
- Naïve Bayes
- SVM
- Random Forest
- Confusion Matrix

- Accuracy Comparison

Suggested Software Tools

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn, mlxtend, Matplotlib, Seaborn
- **Development Environment:** Jupyter Notebook or Google Colab
- **Datasets:** Kaggle, UCI Machine Learning Repository, Data.gov, OpenML

Common Deliverables

1. Problem definition and objectives
2. Dataset description and source
3. Data preprocessing pipeline
4. Exploratory Data Analysis (EDA) and visualizations
5. Implementation of the selected data mining/classification algorithm(s)
6. Performance comparison using evaluation metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC where applicable)
7. Conclusions and recommendations
8. Project report and presentation

These projects align well with the learning outcomes of **Units III and IV** and introduce students to practical applications commonly encountered in data science, business analytics, finance, healthcare, and educational analytics.